Energy-Aware Heterogeneous Multi-Robot Routing with UAV-Assisted Battery Swapping

***Abstract*— Autonomous last-mile delivery systems with homogeneous ground-vehicle fleets face a fundamental energy bottleneck that increases overall mission completion time and total energy consumption. This paper presents a UAV-assisted Battery Management System (BMS) that transforms energy logistics from "come-to-energy" to "energy-comes-to-you" through in-field battery swapping. Our approach integrates baseline routing frameworks with a runtime energy-management layer that replaces unnecessary depot-return detours with strategically deployed UAV battery exchanges. The system is validated through physical prototype development, measuring operational parameters including UAV endurance, and power consumption under realistic constraints. Multi-scenario simulation analysis across varying fleet sizes, task loads, and environment complexities demonstrates consistent performance improvements: 6.48% reduction in total completion time and 10.68% lower energy consumption compared to traditional depot-charging approaches. The system maintains these gains while explicitly enforcing UAV endurance limits and swap operation overhead, making it a scalable solution for large-scale autonomous delivery networks.** https://anonymous.4open.science/r/uav-assisted-lmd-5DB5

I. Introduction

Last Mile Delivery (LMD) systems face a fundamental energy bottleneck: autonomous ground vehicles (UGVs) must abandon active delivery missions to return to charging stations when batteries deplete, reducing fleet productivity by 30–40% [1]. As companies like Serve Robotics and Starship Technologies deploy thousands of delivery robots in urban environments [2, 3], this "return-to-base" energy model creates cascading operational delays and severely limits the effective radius of autonomous last-mile delivery networks [4]. Fixed charging stations require physical docking for energy transfer via tethers [5] or wireless transmission [6, 7], but still force mission interruption. Mobile charging solutions show promise by bringing power to vehicles in the field [8], with recent advances in UAV-based power delivery [9, 10] and battery swapping systems [11] demonstrating the potential for seamless energy replenishment. However, existing approaches lack integration with dynamic task allocation in multi-robot delivery systems.

This paper introduces a UAV-assisted Battery Management System (BMS) that transforms energy logistics from "come-to-energy" to "energy-comes-to-you". Our system deploys drones as autonomous energy couriers that locate delivery vehicles in the field and perform rapid battery swaps without mission interruption, enabling continuous UGV operations through real-time energy management. Figure 1 illustrates the integrated LMD-BMS coordination that maintains fleet productivity through strategic UAV-assisted battery resupply.

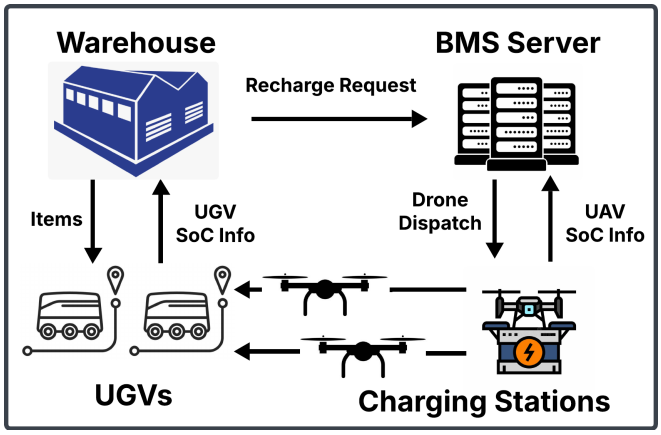


Fig. 1: Integration of the last-mile delivery (LMD) and UAV-assisted Battery Management System (BMS) systems. LMD system (left) handles deliveries using UGVs, while BMS system (right) monitors and performs battery delivery. The warehouse monitors UGV battery levels and requests drone-dispatched battery replacements from the BMS server, ensuring uninterrupted operations through charging stations.

The key contributions of this work are as follows:

- A hybrid optimization framework that integrates EVRP baseline planning with UAV-assisted battery management, enabling continuous delivery operations while reducing mission interruptions.
- Physical prototype development of autonomous battery exchange systems with measured operational parameters (swap timing, UAV endurance, power consumption) that validate simulation models under realistic constraints.
- Multi-scenario simulation analysis demonstrating consistent performance improvements of 6.48% reduction in operational time and 10.68% lower energy consumption compared to traditional depot-charging approaches.

The remainder of this paper is organized as follows. Section II reviews related work in vehicle routing and UAV–UGV coordination. Section III presents the system design, including hardware and software components. Section IV introduces the baseline allocation and routing under energy constraints. Section V details the battery-aware auxiliary layer for depot-free recharging. Section VI describes the simulation environment, performance metrics, and experimental design. Finally, Section VII concludes the paper and outlines future research directions.

II. Related Work

LMD systems require sophisticated optimization approaches balancing routing efficiency with operational constraints, including delivery deadlines, customer time win-

dows, and vehicle capacity limitations. Traditional combinatorial optimization methods derived from the Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP) [12] form the foundation of these systems, typically implemented using Mixed-Integer Programming (MIP) solvers such as Gurobi [13]. Research demonstrates substantial efficiency gains when delivery schedules align with customer availability patterns [14].

Electric Vehicle Routing Problem (EVRP) [15] is the closest to LMD as it involves multiple vehicles with fuel constraints. [16] is the first paper in which they consider a vehicle routing problem involving alternative fuel vehicles (including EVs) that have limited travel range and must re-charge during routing. [17] studied an EVRP with time windows and charging stations (E-VRPTW) and partial recharges. [18] propose a model that is able to consider several alternative-fuel vehicles by adopting different driving ranges, refueling times, and availability of refueling stations. To solve the model, an efficient Variable Neighborhood Search (VNS) heuristic is developed [19].

However, these optimization approaches fundamentally assume that vehicles must return to fixed charging infrastructure when energy depletes, inherently limiting operational radius and introducing the mission interruption problem we seek to address.

Autonomous robotic systems extensively explore battery recharging and swapping mechanisms, categorized into fixed and mobile paradigms [20]. Fixed stations require vehicles to navigate to predetermined locations for power transfer through tethered connections, recharge pins, or wireless transmission systems [5] [6] [7]. Mobile systems offer greater operational flexibility by enabling energy transfer directly at operational sites, with applications ranging from UAVs powering ground robots to airborne power distribution platforms [8] [9] [10]. Battery swapping minimizes operational downtime through rapid replacements [4], with notable implementations including mid-air UAV swapping using specialized Flying Hot-Swap Battery drones [11].

UAV-assisted path planning significantly improves UGV energy efficiency [21], while robust positioning algorithms enhance navigation in complex urban environments [22]. Multi-agent architectures successfully address Vehicle Routing Problems [23], fuel-constrained UAV-UGV route optimization [24], and comprehensive cooperative planning frameworks [25]. While this body of work demonstrates the potential of UAV-UGV coordination, existing systems primarily focus on exploration and surveillance applications rather than the time-sensitive, customer-facing demands of commercial delivery services. Works like [24] use UAVs as the primary vehicles while UGVs act as supporting role for recharging. While this might work for surveillance applications, UGVs are still used as primary for LMD.

III. System Design and Hardware

The proposed system addresses the LMD problem using a fleet of autonomous ground vehicles (UGVs) and unmanned aerial vehicles (UAVs) coordinated through a centralized Battery Management System (BMS). The objective is to ensure that delivery tasks are completed efficiently while mitigating downtime due to limited battery capacity. The design is modular: the baseline delivery framework allocates and routes tasks among UGVs, while the auxiliary BMS framework overlays energy-aware decision-making through UAV-assisted recharging.

A. Actors and Roles

The system comprises three classes of actors:

- **UGVs (Delivery Agents):** Primary task executors responsible for package delivery. Each UGV is energy-limited and must either return to a depot for recharging or receive in-field assistance to remain operational.
- **UAVs (Energy Agents):** Secondary agents responsible for transporting fully charged batteries to UGVs. They enable UGVs to continue their missions without detours to charging stations.
- **BMS (Coordinator):** A centralized controller that monitors battery levels, assigns tasks, and decides whether a UGV should (i) proceed with its next delivery, (ii) request a UAV-delivered battery swap, or (iii) return to the warehouse.

B. Prototype Hardware Implementation

We built a modular battery-exchange prototype to validate feasibility. UGVs are powered by 12 V nickel iron batteries, providing robust and durable energy storage suitable for autonomous ground vehicle operations. The carrier UAV is a lab-built hexacopter using a custom Samsung 25R 18650 pack (4s4p, 14 V, 10 Ah, ≈140 Wh). For delivery, the UAV transports Turnigy Nano-Tech 2200 mAh 3S LiPo packs, which are deposited into the UGV's tray during swaps.

The hardware consists of a 3D-printed docking funnel mounted on the UGV for UAV landing guidance, and a UGV tray with alignment fins and ultrasonic sensors to ensure proper UAV placement (Fig. 2). This low-cost, modular design enables autonomous docking and battery exchange.

From the prototype we logged parameters that directly inform the simulation models: (i) UAV hover and cruise power with/without payload, (ii) endurance limits based on the custom 18650 pack, (iii) swap operation time (approach, alignment, exchange), and (iv) probability of reattempt after a failed alignment. These measurements ground the UAV energy model and swap timing used in our large-scale experiments.

C. Battery Management and Control System

The BMS overlays an energy management layer onto the baseline framework. Each UGV continuously reports its state of charge (SOC), and the BMS applies a simple decision logic:

- **Proceed:** If the SOC suffices for the next delivery and eventual depot return, the UGV continues its planned route.
- **Resupply:** If the SOC is insufficient but recoverable via UAV support, the BMS dispatches the nearest UAV to



(a) (b)

Fig. 2: (a) UAV docked on the exchange system. (b) Battery-exchange tray installed on a UGV. The exchange system incorporates a custom 3D-printed, slider-based mechanism for autonomous battery replacement.

deliver a fresh battery, enabling the UGV to continue without interruption.

- **Return:** If neither option is feasible, the UGV is instructed to return to the depot for recharging.

The BMS integrates battery monitoring, health management, and intelligent dispatch coordination. Standard techniques including Open Circuit Voltage (OCV), Coulomb counting, and Extended Kalman Filtering (EKF) track the state of charge (SOC), health (SOH), charge cycles, temperature, and effective capacity of each battery. The nickel iron batteries offer superior longevity and deep discharge tolerance, making them ideal for intensive delivery operations. When a UGV's SOC drops below threshold, the BMS dispatches the nearest available UAV carrying a replacement pack, accounting for UAV energy reserves, proximity, and task urgency. Degraded batteries are automatically flagged and removed from circulation to ensure fleet reliability. At scale, optimization routines coordinate UAV availability with UGV energy demands, dynamically prioritizing swaps to minimize idle time and maximize throughput. All simulation parameters for costs, and operational constraints are derived from prototype measurements, providing a realistic energy management framework for the experiments.

IV. Baseline Allocation and Routing

To provide a rigorous and battery-aware benchmark for the proposed UAV-assisted Battery Management System (BMS), we adopt the Electric Vehicle Routing Problem (EVRP) formulation as a baseline. Fundamentally, the EVRP can be viewed as an extension of the Multiple Traveling Salesman Problem (mTSP) [26], which the simplest form of VRP where vehicles have infinite range. EVRP, in contrast, has range and refueling constraints. In EVRP, routing and recharging decisions are optimized jointly within a unified mathematical model. Vehicle sequencing, station visits, and recharge quantities are determined endogenously, ensuring that energy feasibility is enforced directly through the optimization constraints rather than through external rules.

A. Problem Formulation

Let $G = (V, E)$ be a directed graph with depot $0 \in V$ and task nodes $N := V \setminus \{0\}$. Let $\mathcal{M} = \{1, \ldots, m\}$ denote the set of homogeneous electric vehicles. Let $\tau_{ij} \geq 0$ be the travel cost associated with arc $(i, j) \in E$, representing both travel time and equivalent energy consumption. The parameter R is the fixed battery range. The decision variables are $x_{kij} \in \{0, 1\}$, equal to 1 if vehicle k traverses arc (i, j), and a scalar $T \geq 0$ representing the makespan. The continuous variable $e_i \geq 0$ tracks the remaining battery energy of a vehicle upon arriving at task node $i \in N$, with the depot departure energy defined as a constant $e_0 = R$.

$$\min \quad T$$

$$\text{s.t.} \quad \sum_{k \in \mathcal{M}} \sum_{i \in V \setminus \{j\}} x_{kij} = 1 \quad \forall j \in N \tag{1}$$

$$\sum_{j \in V \setminus \{i\}} x_{kij} - \sum_{j \in V \setminus \{i\}} x_{kji} = 0 \quad \forall i \in V, k \in \mathcal{M} \tag{2}$$

$$\sum_{j \in N} x_{k0j} \leq 1 \quad \forall k \in \mathcal{M} \tag{3}$$

$$\sum_{(i,j) \in E} \tau_{ij} x_{kij} \leq T \quad \forall k \in \mathcal{M} \tag{4}$$

$$e_j \leq e_i - \tau_{ij} + (R + \tau_{ij})\Big(1 - \sum_{k \in \mathcal{M}} x_{kij}\Big) \quad \forall i \in V, j \in N, i \neq j \tag{5}$$

$$e_i \geq \tau_{i0} \sum_{k \in \mathcal{M}} x_{ki0} \quad \forall i \in N \tag{6}$$

$$x_{kij} \in \{0, 1\} \quad \forall (i, j) \in E, k \in \mathcal{M} \tag{7}$$

$$e_i \geq 0 \quad \forall i \in N \tag{8}$$

$$T \geq 0 \tag{9}$$

The objective minimizes the makespan T. Constraint (1) guarantees that every task node is visited exactly once. Constraint (2) enforces standard flow conservation at all nodes for each vehicle $k \in \mathcal{M}$. Constraint (3) limits each vehicle to at most one departure from the depot. Constraint (4) defines the makespan by bounding T from below by the total travel time accumulated by each individual vehicle. Constraint (5) captures the battery dynamics using Miller-Tucker-Zemlin logic: it deducts the energy τ_{ij} when an arc is traversed, eliminates sub-tours, and uses a Big-M term of $(R + \tau_{ij})$ to relax the bound when the arc is not traversed. Constraint (6) guarantees that the vehicle arriving at its final task node i retains enough energy to return to the depot. The final three constraints establish the binary and non-negativity domains for the decision variables.

B. Methodology

The Electric Vehicle Routing Problem with Time Windows (EVRPTW) [17] and its partial recharge variant (EVRPTW-PR) are addressed by first formulating the problem as a $0-1$ mixed integer linear program. Because large instances of this problem are computationally intractable, an Adaptive Large Neighborhood Search (ALNS) algorithm is utilized to efficiently find high-quality solutions.The ALNS methodology is built on a destroy-and-repair framework that continuously modifies and improves a routing solution.

1) Initial Solution Construction: A feasible starting solution is first created using an iterative insertion heuristic. Candidate customers are evaluated, and the one that increases the total route distance the least is inserted while strictly

maintaining time window and battery capacity constraints. If a customer cannot be reached due to a low battery, a recharging station is added to the route. If the customer still cannot be accommodated due to capacity or time constraints, a new route (vehicle) is opened.

2) Destroy Phase (Removal Algorithms): At each iteration, a portion of the current solution is destroyed by removing a certain number of customers and recharging stations from their routes. Several mechanisms are adaptively selected:

- Customer Removal (CR): Standard methods like Random, Worst-Distance, Worst-Time, Demand Based, and Time Based removal are utilized, as well as Route Removal methods (Random Route Removal and Greedy Route Removal) that eliminate entire routes to reduce the fleet size. Specialized EV mechanisms are also employed: Remove Customer with Preceding Station (RCwPS) and Remove Customer with Succeeding Station (RCwSS), which efficiently eliminate unnecessary station visits when a nearby customer is removed.
- Station Removal (SR): Because recharging stations are critical components, they are periodically removed or swapped. Specialized SR algorithms include Worst Charge Usage Station Removal, which removes stations visited by EVs with already high battery levels, and Full Charge Station Removal, which promotes partial recharging strategies by randomly removing instances of fully charged EVs.

3) Repair Phase (Insertion Algorithms): After the solution is partially destroyed, insertion algorithms repair it by placing the removed customers and newly needed stations back into the routes.

- Customer Insertion (CI): Customers are re-inserted using algorithms like Greedy Insertion, Regret-2, Regret-3, Time Based Insertion (minimizing added route time), Zone Insertion (focusing on a specific geographical zone).
- Station Insertion (SI): SI algorithms are used to repair routes when removing stations or adding customers makes a route battery-infeasible. The first customer where the EV arrives with a negative battery level is identified, and an available station is strategically inserted beforehand. Mechanisms include Greedy Station Insertion, Greedy Station Insertion with Comparison, and Best Station Insertion.

4) Adaptive Selection and Acceptance Criterion: The removal and insertion operators are chosen dynamically throughout the process based on past success. Operators that successfully improve the solution are given a higher probability of being selected in subsequent iterations. Finally, after a new solution is generated, it is evaluated and either accepted or rejected based on a Simulated Annealing (SA) criterion, which allows for the occasional acceptance of worse solutions to escape local optima.

C. Adaptation to the LMD Setting

The EVRPTW-PR formulation described above is adapted to reflect the structural characteristics of the last-mile delivery (LMD) environment considered in this study. Two key simplifications are introduced.

In the LMD setting considered here, delivery tasks do not impose service time windows. Consequently, all temporal constraints associated with customer service start times are removed from the model. Specifically, the service time variables τ_i and their associated constraints enforcing $a_i \leq \tau_i \leq b_i$ are eliminated. As a result, feasibility is governed solely by routing, vehicle capacity, and battery dynamics constraints. All customer nodes are assumed to have identical demand. The uniform-demand assumption removes heterogeneity in customer load requirements while preserving the vehicle capacity constraint.

Under these adaptations, the resulting optimization problem remains a capacitated EVRP with partial recharging on a directed network. The objective continues to minimize total travel distance, and the energy feasibility constraints remain unchanged. The model thus retains the coupled routing–recharging decision structure of the original EVRPTW-PR formulation while reflecting the operational characteristics of the LMD environment studied in this work.

V. Depot-Detour Reduction via Segment-VRP with Distance Minimization

In the baseline EVRP formulation, tasks are clustered among UGVs and each cluster is routed with a TSP heuristic under battery range constraints. Whenever a UGV lacks sufficient energy to continue to its next assigned task, the baseline inserts an explicit *return-to-depot* detour for recharging, which inflates total system travel distance and reduces fleet efficiency. We address this limitation by deploying UAVs from distributed charging stations to deliver batteries to strategically selected swap points, leading to a re-optimization that we call the *Discretized Segment-VRP with Distance Minimization*.

A. BMS Infrastructure Setup

The Battery Management System (BMS) operates through distributed charging stations, each equipped with UAVs, battery stockpiles, and autonomous dispatch capabilities. The BMS monitors UGV battery levels and dispatches the nearest UAV when state of charge drops below operational thresholds, eliminating depot returns through in-field battery swaps.

Figure 3 illustrates how UAV-assisted battery swaps reduce fuel requirements (left) and the optimization framework for selecting strategic swap points along route segments (right).

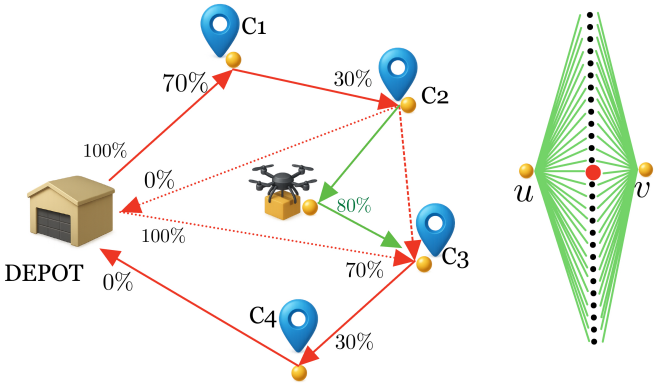


Fig. 3: BMS-assisted reduction of depot detours. Left: UAV-delivered battery swaps (green) replace costly ground vehicle returns to the depot (red), reducing travel distance and energy consumption. Right: a delivery segment between nodes u and v is discretized into candidate swap points, with midpoint analysis guiding the selection of cost-minimizing rendezvous locations.

B. Problem statement.

We are given line segments $S = \{1, \ldots, |S|\}$ where segment i has endpoints $a_i, b_i \in \mathbb{R}^2$ and midpoint $m_i = \frac{a_i+b_i}{2}$. Each segment has candidate swap points $V_i = \{p_{i1}, \ldots, p_{iM_i}\} \subset [a_i, b_i]$. There are K UAV charging stations $\{s_1, \ldots, s_K\} \in \mathbb{R}^2$.

A feasible solution selects one swap candidate per segment and constructs UAV routes visiting every segment exactly once. The objective minimizes UAV flight distances $c(u, v) = \|u - v\|_2$ and midpoint deviations $d(p_{ij}) = \|p_{ij} - m_i\|_2$:

$$\min \quad \sum_{k=1}^{K} \sum_{u,v} c(u, v)\, x_{uv}^k + \beta \sum_{i=1}^{|S|} \sum_{j=1}^{M_i} d(p_{ij})\, y_{ij}, \qquad (10)$$

where $y_{ij} = 1$ if swap candidate p_{ij} is chosen, and $x_{uv}^k = 1$ if UAV k traverses arc (u, v).

Algorithm 1 On-Demand UAV Deployment for Battery Swapping

Input: UGV states $\{SOC_i, pos_i, route_i\}$; charging stations $\{s_k\}$; threshold τ
Output: UAV dispatch decisions and swap locations
while *system operational* **do**
  foreach *UGV i with $SOC_i < \tau$* **do**
    Predict energy requirement for remaining route E_{req}
    if *$SOC_i < E_{req}$ and swap beneficial* **then**
      Find nearest available charging station s^* with UAV Determine optimal swap location p^* on current route Dispatch UAV from s^* to p^* with battery Schedule UGV to rendezvous at p^* Update UAV and station availability
    end if
  end foreach
  Monitor ongoing swaps and update system state
end while

C. Computational complexity.

The Segment-VRP with Distance Minimization is NP-complete. NP membership follows from polynomial-time verification of solutions. NP-hardness is proven by reduction from TSP: construct a single UAV instance where each vertex becomes a segment with one candidate point.

D. MILP formulation.

Let $V = \bigcup_{i \in S} V_i$ denote all candidate swap nodes. The flow-based MILP is:

$$\min \quad \sum_{k=1}^{K} \sum_{u,v} c(u, v)\, x_{uv}^k + \beta \sum_{i=1}^{|S|} \sum_{j=1}^{M_i} d(p_{ij})\, y_{ij}$$

$$\text{s.t.} \quad \sum_{j=1}^{M_i} y_{ij} = 1 \quad \forall i \in S, \qquad (11)$$

$$\sum_{k=1}^{K} \sum_{u} x_{u,p_{ij}}^k = y_{ij} \quad \forall i \in S, j = 1, \ldots, M_i, \qquad (12)$$

$$\sum_{k=1}^{K} \sum_{v} x_{p_{ij},v}^k = y_{ij} \quad \forall i \in S, j = 1, \ldots, M_i, \qquad (13)$$

$$\sum_{v \in V} x_{s_k,v}^k \le 1, \quad \sum_{u \in V} x_{u,e_k}^k \le 1 \quad \forall k = 1, \ldots, K, \qquad (14)$$

$$\sum_{u \in V \cup \{s_k\}} x_{u,v}^k = \sum_{w \in V \cup \{e_k\}} x_{v,w}^k \quad \forall k, \forall v \in V, \qquad (15)$$

$$\text{subtour elimination constraints}, \quad x_{uv}^k, y_{ij} \in \{0, 1\}.$$

Constraints (11)-(13) ensure segment coverage and visitation, (14) handles charging station departures/returns, and (15) maintains flow conservation.

Reduction from depot-return gaps Given consecutive tasks (a_i, a_{i+1}) with depot detour distance $d_{\text{dep}}(i) = d(a_i, w) + d(w, a_{i+1})$, we construct surrogate segment S_i using the perpendicular bisector of $\overline{a_i a_{i+1}}$ with endpoints satisfying $d(a_i, p_i^-) + d(p_i^+, a_{i+1}) = \alpha \cdot d_{\text{dep}}(i)$ where $0 < \alpha < 1$ ensures distance reduction.

Outcome. The system provides real-time UAV deployment that replaces depot detours $(a_i \to w \to a_{i+1})$ with battery swaps $(a_i \to p^* \to a_{i+1})$, reducing total distance while maintaining operational feasibility.

VI. Simulation Results

A. Simulation Setup

Simulations were conducted in a ROS Noetic (Ubuntu 20.04) environment with Gazebo 11.15.1 providing physics and rendering. The last-mile delivery (LMD) environment is modeled as a 2D maze, where distances correspond to shortest-path traversal over the grid rather than Euclidean metrics. UGVs execute delivery tasks while UAVs (in BMS-enabled variants) perform in-field battery swaps. UAV flight power, endurance limits, and swap operation times are parameterized using measurements obtained from the physical prototype, including wind-aware power modeling and fixed swap overhead.

To evaluate robustness and scalability, the four LMD models are tested across five key parameters: (i) number of UGVs, which controls fleet size and coordination load; (ii) number of delivery tasks, representing workload intensity;

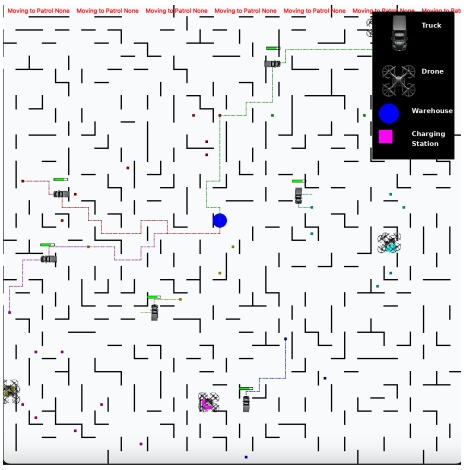


Fig. 4: Snapshot of the simulation environment used for last-mile delivery tasks.

(iii) electric vehicle (EV) range, defining the maximum travel distance between recharge events; (iv) maze loop percentage, which governs environment connectivity and path redundancy; and (v) maze size, which scales the spatial extent of the operational area. By varying these parameters systematically, we assess performance across fleet scaling, demand scaling, energy constraints, and environmental complexity.

B. Performance Metrics

Performance is evaluated using two primary metrics: makespan and total energy consumption.

a) Makespan.: Makespan is defined as the total time required to complete all delivery tasks, measured as the maximum completion time across all vehicles:

$$T = \max_{i \in \mathcal{T}} C_i,$$

where C_i denotes the completion time of task i. This metric captures overall fleet efficiency, including routing time, depot returns (if any), UAV swap operations, and associated coordination delays.

b) Total Energy Consumption.: Total energy consumption aggregates the distance-weighted energy usage of all active agents:

$$E = \sum_{m \in \mathcal{M}} \alpha \, d_m^{\text{UGV}} + \sum_{u \in \mathcal{U}} \beta \, d_u^{\text{UAV}},$$

where d_m^{UGV} and d_u^{UAV} denote the total travel distances of UGV m and UAV u, respectively. The coefficients α and β represent per-distance energy costs derived from the calibrated vehicle and UAV power models. In non-BMS configurations, the second term is zero.

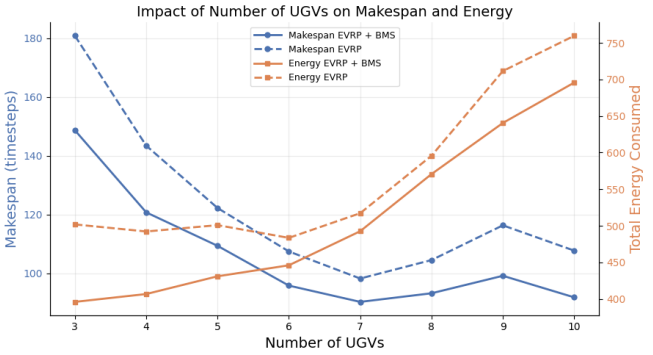


Fig. 6: Comparison of BMS (solid lines) vs. non-BMS (dashed lines), showing makespan (blue, left axis) and energy consumption (orange, right axis) across different UGV counts.

C. Simulation Scenarios

The simulation setup allows tuning of multiple parameters such as UGV count, task count, UGV range, number of base stations, and maze connectivity. Since exhaustively enumerating all combinations is infeasible, we focused on three dominant factors: (i) task load, (ii) UGV Range, (iii) maze complexity, and (iv) fleet size. We randomize the size of the maze between 35 and 50. This ensures tractability while still revealing the effect of BMS on the most influential aspects of LMD.

We evaluate the performance of the current best benchmark in LMD (EVRP) against EVRP augmented with BMD. For the EVRP+BMS configuration, recharge station visits present in the original EVRP solution are removed prior to execution. Since UAV-assisted swapping provides dynamic

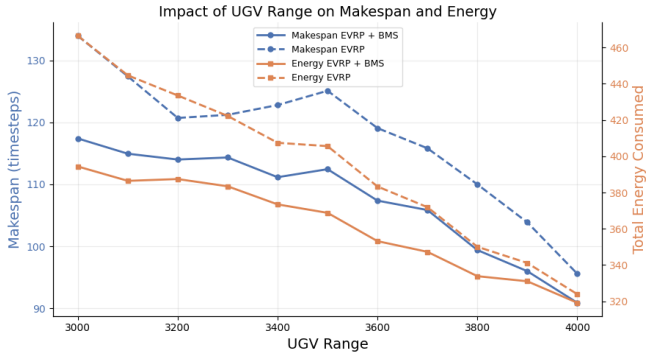


Fig. 5: Comparison of BMS (solid lines) vs. non-BMS (dashed lines), showing makespan (blue, left axis) and energy consumption (orange, right axis) across different UGV battery range.

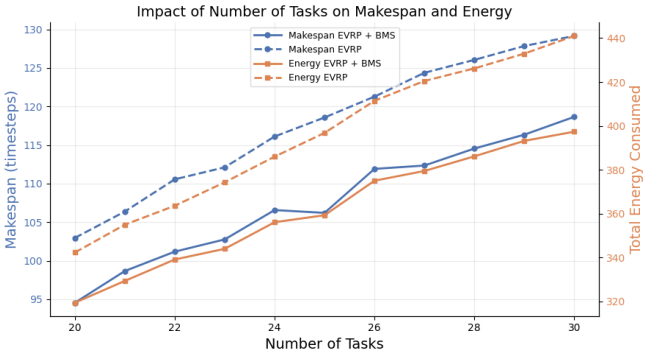


Fig. 7: Comparison of BMS (solid lines) vs. non-BMS (dashed lines), makespan (blue, left axis) and energy consumption (orange, right axis) across different task counts.

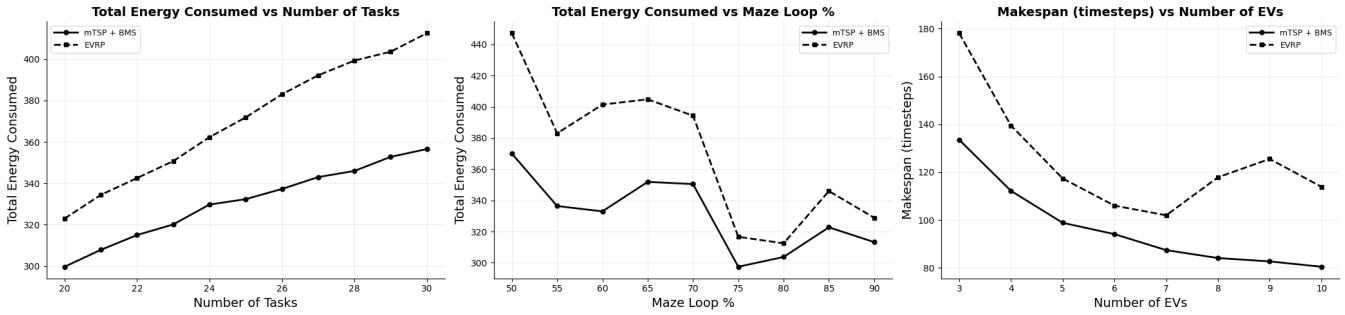


Fig. 8: Energy consumption and makespan of mTSP+BMS and EVRP, across varying task loads, maze connectivity (λ), and fleet sizes.

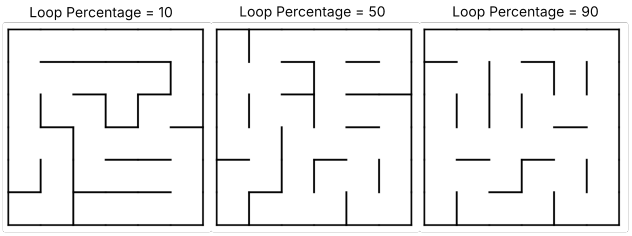


Fig. 9: Maze structures with varying loop percentages: 10%, 50%, and 90%. Higher λ values increase connectivity and reduce dead ends.

energy support, it is no longer meaningful to enforce pre-planned depot recharge stops when the vehicle's state-of-charge remains operational. Instead, energy feasibility during execution is enforced using Algorithm 1 (Range Enforcement), which determines depot returns only when strictly necessary.

1) EV Range: As EV range increases, as seen in Figure 5, both EVRP and EVRP+BMS show a steady reduction in makespan and total energy consumption. EVRP+BMS consistently outperforms EVRP across all range values, maintaining lower makespan and energy throughout. The difference between the two models is more pronounced at lower EV ranges and gradually narrows as range increases. At higher ranges, both curves begin to flatten, indicating diminishing improvements with additional range. The narrowing gap reflects the reduced need for UAV-assisted battery swaps as UGVs are able to complete longer segments without

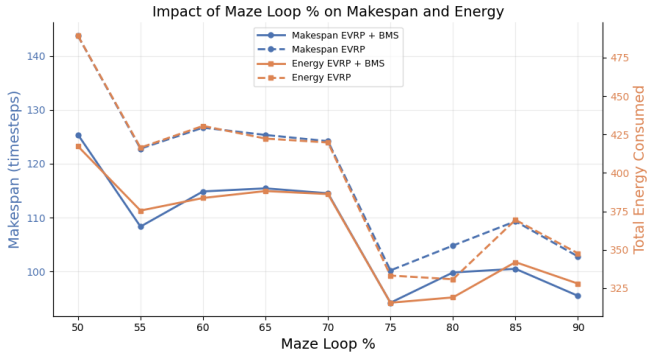


Fig. 10: Comparison of BMS (solid lines) vs. non-BMS (dashed lines), showing makespan (blue, left axis) and energy consumption (orange, right axis) across maze complexities.

external energy support.

2) Number of UGVs: Figure 6 shows the impact of the number of EVs on makespan and total energy consumption for EVRP and EVRP+BMS. As the fleet size increases, makespan decreases for both models, with EVRP+BMS consistently achieving lower values across all fleet sizes. EVRP+BMS also maintains lower energy consumption than EVRP throughout. Overall, EVRP+BMS demonstrates consistent improvements in makespan while retaining an energy advantage across varying fleet sizes.

3) Tasks: Figure 7 illustrates the effect of increasing the number of tasks on makespan and total energy consumption for EVRP and EVRP+BMS. As the task count increases from 20 to 30, both makespan and energy consumption rise monotonically for the two models. EVRP+BMS consistently achieves lower makespan across all task volumes, with a relatively stable gap as workload increases. A similar pattern is observed for total energy consumption, where EVRP+BMS maintains uniformly lower energy usage compared to EVRP. The trends for both models are approximately linear, indicating proportional scaling of operational cost with task load, while preserving the performance advantage of the BMS-enabled system.

4) Maze Complexity: We evaluated the effect of environment connectivity, modeled by the loop percentage λ. Low λ creates tree-like mazes with many dead ends, while high λ yields highly interconnected layouts with multiple redundant paths (Figure 9).

Figure 10 shows the impact of environment connectivity, modeled by the loop percentage λ, on makespan and total energy consumption for EVRP and EVRP+BMS. As λ increases from 50% to 75%, both models exhibit a noticeable reduction in makespan and energy consumption, reaching their lowest values near $\lambda = 75\%$. Beyond this point, the metrics show mild fluctuations but remain lower than those observed in sparsely connected environments. Across all connectivity levels, EVRP+BMS consistently maintains lower makespan and total energy consumption compared to EVRP, with the performance gap preserved in both low- and high-connectivity regimes.

D. Discussion

An important observation from the results is that augmenting a simple routing baseline with BMS can outperform more

sophisticated battery-aware routing models. In particular, the results show that mTSP+BMS consistently performs better than EVRP across the evaluated scenarios. This trend is visible in both the energy and makespan comparisons shown in Figure 8. Across increasing task counts and varying environment connectivity, mTSP+BMS maintains lower total energy consumption than EVRP, while also achieving lower makespan as fleet size increases.

This result highlights a key practical advantage of the proposed framework. EVRP jointly optimizes routing and recharge planning, which significantly increases computational complexity. In contrast, mTSP is a simpler formulation that ignores battery constraints during planning and can be solved much faster. By shifting energy management to execution through UAV-assisted swaps, BMS allows battery feasibility to be maintained dynamically without embedding recharge constraints into the routing optimization. As a result, a lightweight planner such as mTSP, when combined with BMS, can outperform EVRP while avoiding the computational overhead of solving a complex battery-constrained routing problem.

VII. Conclusion

This paper presents a UAV-assisted Battery Management System that addresses the inefficiency in homogeneous autonomous delivery systems caused by depot charging requirements. Our approach integrates Segment-VRP optimization with physical prototype validation, demonstrating 6.48% reduction in operational time, and 10.68% lower energy consumption across diverse scenarios. Future work will focus on predictive energy management using machine learning, multi-objective optimization for complex operational requirements, and real-world deployment studies to validate commercial viability in urban environments.

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